Prior Information-Guided Semi-Supervised Semantic Segmentation of Remote Sensing Images

Xiaorong Gan, Wenting Li, Yongjun Zhang, Wei Long, Yujie Lu, and Ziyang Chen, *Student Member, IEEE*

***Abstract*— Remote sensing semantic segmentation has broad applications in critical fields such as urban planning and environmental monitoring. Semi-supervised semantic segmentation methods tackle the time and cost challenges of annotating remote sensing images by using abundant unlabeled data to optimize the model. Many existing semi-supervised methods generate pseudo-labels by inputting unlabeled data into the network and then using these labels to progressively improve the model's segmentation performance. However, the imbalanced class distribution and complex spatial layout of remote sensing images may result in unreliable pseudo-labels generated by the networks. To address this issue, we propose a prior information-guided network (PGNet) that integrates a prior feature-guided module (PFGM) for reliable feature priors and a prior edge extraction module (PEEM) to provide edge prior information. By leveraging the lightweight segment anything model (SAM), the PGNet enhances the ability to extract valuable information from unlabeled data, enabling the generation of more reliable pseudo-labels. The PFGM employs symmetric cross-attention to more balancedly integrate the SAM prior features with the semantic features from the encoder. Additionally, the PEEM employs the SAM to extract reliable edge priors from unlabeled data, guiding the feature maps during training for more accurate segmentation boundaries. We conducted extensive experiments on the ISPRS Vaihingen, ISPRS Potsdam, and LoveDA datasets. The results demonstrate that our PGNet outperforms existing state-of-the-art (SOTA) semi-supervised methods.**

***Index Terms*— Prior information, remote sensing, segment anything model (SAM), semantic segmentation, semi-supervised.**

I. Introduction

THE rapid advancement of satellite technology and drones has resulted in a significant increase in the number of remote sensing images. This growth has provided abundant data resources for a wide range of application fields [1], such as environmental monitoring [2], [3], [4], road extraction [5], [6], land management [7], and resource exploration [8]. Semantic segmentation is an important task in remote sensing image analysis [9] that aims to classify each pixel in an image, which helps in the precise identification and analysis of geospatial objects and features.

Fully supervised semantic segmentation methods for remote sensing images are highly dependent on a substantial amount of labeled data [10], [11], [12]. However, annotating high-resolution remote sensing images is both time-consuming and labor-intensive. Additionally, the vast geographic coverage and diverse object categories in remote sensing scenes further complicate the task of obtaining high-quality and accurate annotations. In response to these challenges, researchers have increasingly focused on reducing the models' reliance on labeled data. Among the explored methodologies, semi-supervised learning has gained prominence as a promising approach [13], [14], [15], [16]. It aims to develop a robust network model by leveraging a limited set of labeled samples alongside a substantial volume of unlabeled data.

Current mainstream semi-supervised semantic segmentation methods can be broadly divided into two primary categories. The first category consists of consistency regularization methods based on pseudo-labels [14]. These methods are grounded in the idea that the output of the model should remain consistent for different perturbations of the same input. To achieve this, many approaches [17], [18], [19], [20], [21] apply various levels of perturbations to unlabeled data and compute a consistency loss between the outputs of differently perturbed inputs to guide training. The second category comprises self-training methods [22], [23], which typically involve training a model on labeled data first and then using the trained model to generate predictions for unlabeled data. High-confidence predictions are treated as pseudo-labels and incorporated into the training set to further enhance the model's performance.

Although existing methods have considerably enhanced the accuracy of semi-supervised semantic segmentation, the challenge of obtaining more reliable pseudo-labels remains. The unreliability of pseudo-labels arises from the fact that they are generated by the model itself rather than by ground-truth (GT) labels and thus often contain inherent uncertainty. This issue becomes particularly pronounced when the model encounters complex or unfamiliar boundary regions. To address this limitation, we incorporate reliable prior knowledge into the model

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to improve the accuracy of pseudo-label generation. This improvement enables more effective utilization of the abundant unlabeled data, thereby enhancing the overall performance of the model.

In recent years, the introduction of the segment anything model (SAM) [24] has made a profound impact on numerous computer vision tasks. Beyond creating a sensation in the field of image segmentation, the SAM has shown distinct advantages in domains such as image restoration and object detection [25], [26], [27], [28]. The exceptional segmentation performance of the SAM is attributed to its pretraining on large-scale, diverse datasets [25]. These datasets, covering a wide range of scenes, objects, and complex environments, equip the SAM with robust generalization capabilities. As a result, the SAM can effectively identify object boundaries and structures, even in the absence of labeled data. Its universal image encoder captures high-level visual features, such as edges, shapes, and textures, enabling a comprehensive understanding of object attributes. These features are dynamically utilized by the SAM's decoder to generate segmentation masks based on various input prompts. Leveraging this deep understanding of image features, the SAM extracts rich prior information from unseen images, which proves particularly beneficial for leveraging unlabeled data in semi-supervised learning tasks.

The objective of the SAM is to segment anything, rather than to recognize anything. The core functionality of the SAM lies in identifying the boundaries, shapes, and spatial structures of objects, without concern for their specific categories. Its primary task is to generate a valid mask for a given input image and prompt, a process that does not involve explicit semantic supervision. Consequently, the features extracted by the SAM lack semantic information. In contrast, the goal of semantic segmentation is not only to accurately classify pixels but also to assign a category label to each pixel. For this reason, despite the SAM's strong generalization capabilities, it cannot be directly applied to remote sensing image semantic segmentation tasks. Nevertheless, the prior information provided by the SAM can be effectively exploited to enhance the performance of semantic segmentation models.

By leveraging the SAM to extract prior information, we present the PGNet for semi-supervised semantic segmentation of remote sensing images. This prior information enables the PGNet to enhance the reliability of pseudo-labels generated from unlabeled data. Specifically, we utilize the nonsemantic features extracted by the SAM in the initial stage and incorporate them with the semantic features extracted by the semantic segmentation model. This aggregation enhances the model's ability to classify and recognize shared features. Furthermore, to ensure reliable boundary information for unlabeled data, we leverage the efficient segmentation capability of the lightweight SAM [29] to process complex scenes in the unlabeled data. This preprocessing step extracts reliable boundary prior information, which serves as a form of supervision for the model. Extensive experiments were conducted to evaluate the effectiveness of the proposed model guided by prior information. As shown in Fig. 1, the introduction of prior information greatly enhances the model's ability to perceive boundaries. In the visualization, the red box highlights the significant improvements achieved through the incorporation of prior information.

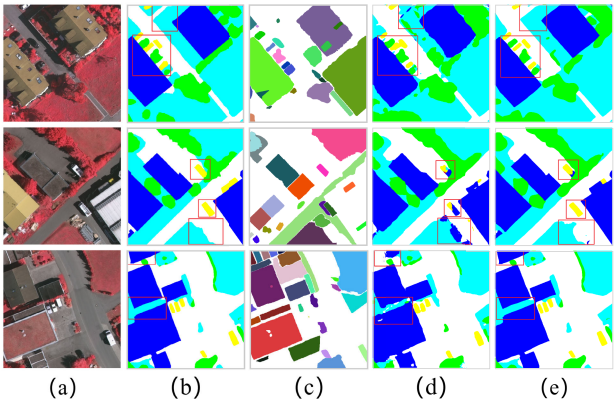


Fig. 1. Performance improvement brought by the incorporation of prior information. (a) Input image. (b) GT. (c) Segmentation result of the unlabeled data obtained through the lightweight SAM, which lacks semantic information. (d) and (e) Visualization results of the baseline model and the PGNet after incorporating prior information, respectively.

In summary, our main contributions are as follows.

1) We propose a prior information-guided network (PGNet) for semi-supervised semantic segmentation of remote sensing images. By integrating prior information from the lightweight SAM, the model is provided with more reliable features and edge priors. This enables the PGNet to better leverage semantic category information from unlabeled image data.
2) We introduce a prior feature-guided module (PFGM) that adopts shared similarity weights and symmetric cross-attention strategies. These strategies ensure that the SAM prior is more balanced with the semantically rich features from the encoder without increasing the computational effort, ensuring equal attention to prior information and semantic information.
3) To enhance the reliability of boundary information in unlabeled data, we design the prior edge extraction module (PEEM) dedicated to the training stage, enhancing the model's perception of object boundaries.

The organization of the remaining parts of this article is as follows. Section II provides a review of the related work. Section III describes the proposed methodology in detail. Section IV presents and discusses the experimental results. Section V concludes this article.

II. Related Work

A. Semantic Segmentation of Remote Sensing Images

1) Fully Supervised Semantic Segmentation: Traditional semantic segmentation methods [30], [31] rely on handcrafted features, are limited to extracting low-level features like color and texture, and fail to capture high-level semantics. In contrast, deep convolutional neural networks (DCNNs) effectively extract deep contextual features, offering significant

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advantages and superior performance in semantic segmentation tasks. Fully convolutional networks (FCNs) [32] were the first CNN models to successfully implement end-to-end pixel-level semantic segmentation, significantly improving accuracy. However, the downsampling operation of FCNs causes the loss of spatial details. To overcome this problem, numerous studies and enhancements [33], [34], [35], [36], [37], [38] have been proposed. Despite this, CNN-based methods, while effective at capturing local features, often struggle with extracting long-range dependencies. Therefore, Transformer-based approaches [39], [40], [41], [42], [43], [44] have been explored for semantic segmentation and have achieved remarkable performance.

In the past few years, researchers have made significant progress in fully supervised semantic segmentation of remote sensing imagery [45], [46]. For instance, the CTMFNet [47] introduces a dual-branch attention fusion module within an encoder–decoder framework to combine local and global information. Subsequently, Sun et al. [48] developed a successive pooling attention module and feature fusion module in their successive pooling attention network. To enhance the feature extraction capability of the self-attention mechanism, the TANet [11] introduces a thresholded attention mechanism, achieving excellent performance with reduced computational cost. The BEDSN [49] employs a dual-stream approach that uses edge detection to address the boundary loss in semantic segmentation. Unlike clustering tasks [50], [51], semantic segmentation typically requires labeled data to achieve optimal performance. However, fully supervised semantic segmentation relies heavily on annotated data, which is both costly and time-consuming, thereby significantly limiting its scalability and applicability. In contrast, semi-supervised semantic segmentation requires only a small amount of labeled data while leveraging large volumes of unlabeled data to extract useful features, thereby enhancing model performance and generalization.

2) Semi-Supervised Remote Sensing Semantic Segmentation: Semi-supervised learning combines a small amount of labeled data with a large quantity of unlabeled data for training, thereby reducing the dependency on labeled data. In its early stages, methods such as generative adversarial networks (GANs) were employed as auxiliary supervision branches to achieve semi-supervised semantic segmentation [17], [52], [53]. More recently, consistency regularization-based methods have gained prominence. These approaches assume that different perturbations applied to the same unlabeled data should yield consistent predictions, leading to a series of advanced techniques [54], [55], [56]. Additionally, self-training methods [57] iteratively refine the quality of pseudo-labels generated for unlabeled data, progressively improving both the pseudo-labels and the model's performance. Some methods [18], [20] further merge self-training with consistency regularization to extract more effective features for enhanced segmentation performance.

Remote sensing images are challenging to annotate due to the irregularity of the objects they contain, making semi-supervised and weakly supervised [58] methods particularly essential. In semi-supervised methods, LPCR [59] proposes a method based on label propagation and contrastive regularization that mitigates the class imbalance between labeled and unlabeled data. MDMCR [60] combines perturbations in low-dimensional spaces and high-dimensional semantic feature spaces to create multidimensional perturbation information, effectively mitigating overfitting and excessive perturbations in low-dimensional spaces. Furthermore, ADEE [61] demonstrates that applying multidisturbance dynamic consistency (MDC) to unlabeled data and aligning multiple predictions from different data augmentations can enrich the perturbation space, helping segmentation models learn more discriminative feature representations. The recent study DWL [62] introduced a decoupled weighted learning framework that reduces the negative impact of incorrect pseudo-labels through decoupled learning, further enhancing segmentation performance.

Although existing studies have advanced semi-supervised semantic segmentation for remote sensing images, they typically rely on the network's own knowledge, resulting in unreliable pseudo-labels during the early training stages. To improve the accuracy of pseudo-labels and enhance the model's ability to learn image boundaries, we develop a semi-supervised semantic segmentation network guided by prior information. This network utilizes prior knowledge to direct feature extraction, thereby increasing the reliability of pseudo-labels throughout the training process.

B. SAM and Its Lightweight Versions

Recently, the emergence of the SAM [24] has marked a significant breakthrough in the field of segmentation. The exceptional capabilities of the SAM can be attributed to two key factors. First, its training on large-scale datasets has enabled the SAM to learn rich feature representations. The diversity in these datasets enhances the model's generalization ability, which has been consistently demonstrated by its remarkable performance across various domains including remote sensing image analysis [25], [26], [27], [63], [64], [65]. Second, the attention mechanism within SAM's encoder plays a crucial role in capturing both global information and fine-grained details, while effectively establishing long-range dependencies. These characteristics endow SAM with strong zero-shot generalization capabilities, allowing it to perform exceptionally well even in complex scenarios. Leveraging SAM's robust zero-shot generalization, we can further utilize it to assist in training with unlabeled data in semi-supervised learning settings, significantly reducing the reliance on labeled datasets.

The SAM has demonstrated exceptional performance across various domains, but a key challenge limits its broader application: its image encoder, based on the Vision Transformer architecture, requires substantial computational resources. To mitigate this issue, several lightweight adaptations of the SAM have been developed. The FastSAM [66] employs a CNN-based architecture to enhance efficiency, while the MobileSAM [67] makes use of architecture distillation to obtain a more lightweight encoder. Additionally, the EfficientSAM [29] uses pretraining on masked images,

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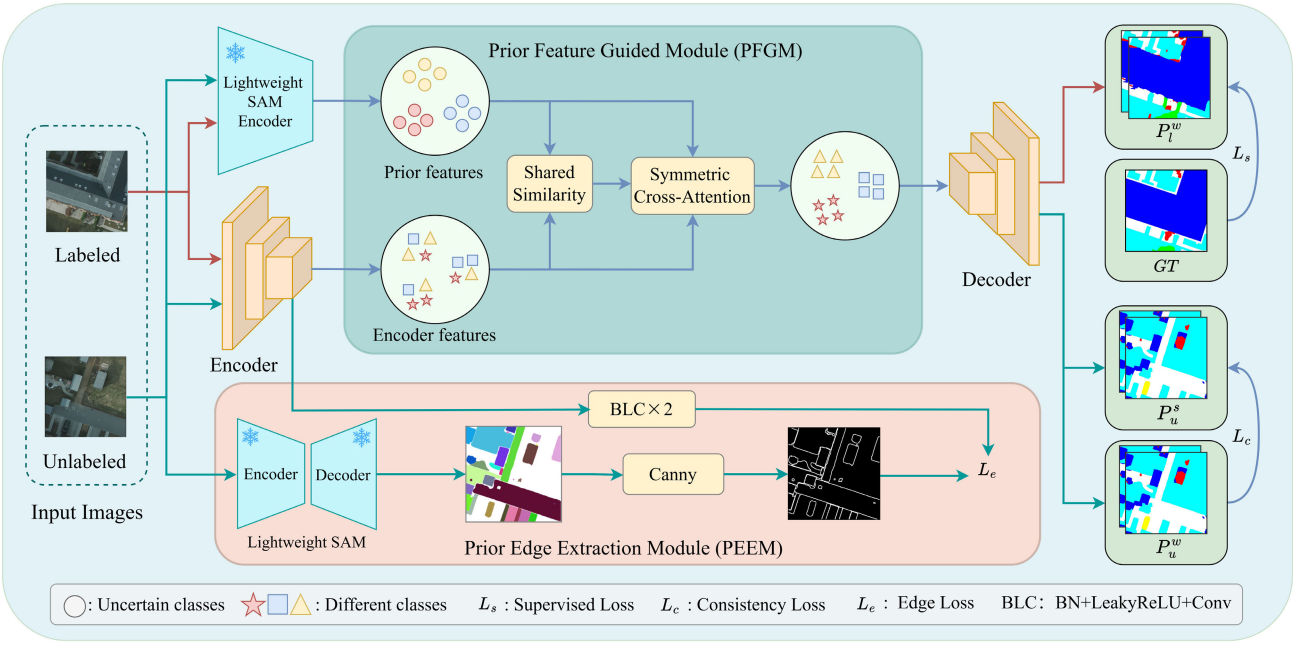


Fig. 2. Overall architecture of the PGNet. The PGNet consists of three main components. The encoder–decoder network extracts semantic features from the input image and restores detailed information for semantic segmentation. The PFGM module integrates prior features from the lightweight SAM with the network encoder's output, enabling more accurate class prediction. To refine the segmentation boundaries, the PEEM module utilizes reliable edge information from SAM to guide the output features of the backbone network. In the figure, P_l^w represents the predicted results of labeled data after weak perturbation; P_u^w and P_u^s represent the predicted results of unlabeled data after weak and strong augmentation, respectively.

enabling it to learn feature reconstruction from the SAM's encoder, thereby achieving efficient visual representation. These lightweight versions significantly expand the scope of the SAM's practical deployment, making it more feasible for resource-constrained scenarios.

Therefore, we introduce using the intermediate features and the segmentation edges extracted by the lightweight SAM as prior information to guide the semi-supervised semantic segmentation of remote sensing images. By combining the prior features of the SAM encoder with the semantic features extracted by the model and designing appropriate modules to deal with these two types of information, the model's feature aggregation ability is enhanced. Moreover, we design an edge-prior loss to supervise the output features of unlabeled data, guiding the model to focus on boundary features, thereby enhancing its ability to recognize the boundaries of target objects.

III. Methodology

A. Overview

We propose the PGNet, a novel framework designed to effectively incorporate prior knowledge from the lightweight SAM foundational model to enhance the semi-supervised semantic segmentation of remote sensing images. As shown in Fig. 2, the PGNet is composed of three key components working seamlessly together. First, the encoder–decoder network serves as the backbone to extract semantic features from the input image and restore detailed information for precise segmentation. Building on this, we design the PFGM module, which enhances the model's feature extraction capabilities by introducing prior features from the SAM encoder through a symmetric cross-attention mechanism. Furthermore, to refine segmentation boundaries, we incorporate the PEEM module, which utilizes reliable boundary information from the SAM to supervise the backbone network's segmentation edge accuracy. This incorporation approach allows the PGNet to achieve superior performance in exploiting prior knowledge for remote sensing image analysis.

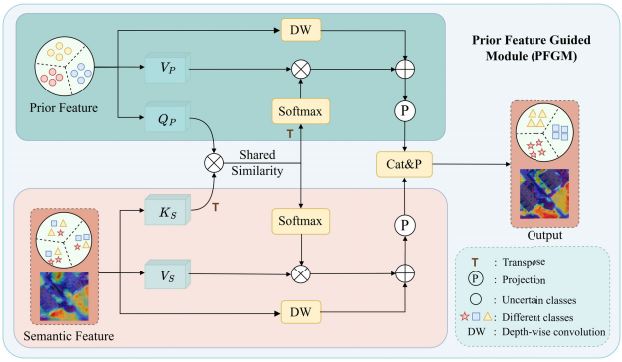


Fig. 3. PFGM. The PFGM receives two input features: one from the output of the EfficientSAM encoder, which aggregates similar features. The other input is the output of the model encoder, which extracts semantic information. The PFGM can make the same semantic information better aggregated.

B. Prior Feature-Guided Module

To effectively utilize the prior feature information provided by the SAM to guide the semantic features extracted by the encoder, we designed the PFGM. As illustrated in Fig. 3, the PFGM employs symmetric cross-attention computations based on shared similarity to better integrate these two

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complementary sources of information. The upper branch represents the reliable prior feature maps extracted from the lightweight SAM encoder. It adopts a lightweight Vision Transformer structure, which can enhance the model's ability to model the global features of the image and remain frozen during training. The lower branch represents the output of the encoder. The network encoder adopts the ResNet-101 structure, which extracts local features by using convolution operations, and its output contains semantically rich features derived from the input data. The SAM primarily captures object boundary features and distinguishes all objects without semantic differentiation, leaving the category information of these features undefined. In contrast, the encoder features encompass detailed semantic information but lack precise structural details. To achieve this complementarity, the PFGM employs a symmetric cross-attention module to facilitate the interaction between these two types of features, enabling a more effective fusion of the semantic richness of the encoder features and the structural clarity of the SAM features.

Let the input image be denoted as X_{in}. After passing through both the network encoder and the encoder of the lightweight SAM, two feature representations containing different types of information are obtained. The formulation is as follows:

$$\mathbf{X}_s = \mathbf{E}_n(\mathbf{X}_{\text{in}}) \tag{1}$$

$$\mathbf{X}_p = \mathbf{E}_{\text{SAM}}(\mathbf{X}_{\text{in}}). \tag{2}$$

The semantic features output by the encoder are denoted as $\mathbf{X}_s \in \mathbb{R}^{C\times N}$, where $N = H \times W$. $\mathbf{E}_n$ represents the network encoder and $\mathbf{E}_{\text{SAM}}$ represents the encoder of the lightweight SAM. To facilitate subsequent attention computation, the output of the SAM is transformed into the same shape, denoted as $\mathbf{X}_p \in \mathbb{R}^{C\times N}$. We employ four distinct projection matrices, $\mathbf{W}_{Qp} \in \mathbb{R}^{C\times C}$, $\mathbf{W}_{Vp} \in \mathbb{R}^{C\times C}$, $\mathbf{W}_{Ks} \in \mathbb{R}^{C\times C}$, and $\mathbf{W}_{Vs} \in \mathbb{R}^{C\times C}$, to derive the corresponding feature matrices $\mathbf{Q}_p$, $\mathbf{V}_p$, $\mathbf{K}_s$, and $\mathbf{V}_s$. These four feature matrices share the same dimensionality, $\mathbb{R}^{C\times N}$. The specific formulas are as follows:

$$\mathbf{Q}_p = \mathbf{W}_{Qp}\ \mathbf{X}_p \tag{3}$$

$$\mathbf{V}_p = \mathbf{W}_{Vp}\ \mathbf{X}_p \tag{4}$$

$$\mathbf{K}_s = \mathbf{W}_{Ks}\ \mathbf{X}_s \tag{5}$$

$$\mathbf{V}_s = \mathbf{W}_{Vs}\ \mathbf{X}_s. \tag{6}$$

Unlike traditional cross-attention mechanisms, our symmetric cross-attention calculates attention weights independently for two different input features. Additionally, it requires only a single matrix multiplication to obtain the attention values for both inputs. The computation is as follows:

$$\mathbf{S}_s = \mathbf{Q}_p\mathbf{K}_s^T \tag{7}$$

$$\mathbf{S}_p = \mathbf{K}_s\mathbf{Q}_p^T = \mathbf{S}_s^T \tag{8}$$

$$\mathbf{A}_s = \text{Softmax}\left(\mathbf{S}_s/\sqrt{d}\right) \tag{9}$$

$$\mathbf{A}_p = \text{Softmax}\left(\mathbf{S}_p/\sqrt{d}\right) = \text{Softmax}\left(\mathbf{S}_s^T/\sqrt{d}\right) \tag{10}$$

where d represents the feature dimension. The Softmax function is applied to perform nonlinear normalization of the similarity scores. First, the prior features $\mathbf{Q}_p$ output by the SAM encoder are projected onto $\mathbf{K}_s^T$ from the network encoder to compute the feature similarity $\mathbf{S}_s$. Then, by transposing the result, we obtain the reverse projection $\mathbf{S}_p$ without any additional matrix computations. This allows us to compute two mutually symmetric similarity results with a single matrix multiplication, from which the respective attention weights are derived. This approach significantly enhances the expressive capacity of the features.

Subsequently, the two obtained attention matrices are multiplied with $\mathbf{V}_s$ and $\mathbf{V}_p$, respectively, to generate the result matrices as linear combinations based on the normalized attention weights. The mathematical formulation is as follows:

$$\mathbf{F}_s = \mathcal{P}\left(\mathbf{A}_p\mathbf{V}_s + \mathcal{D}_{7\times7}(\mathbf{V}_s)\right) \tag{11}$$

$$\mathbf{F}_p = \mathcal{P}\left(\mathbf{A}_s\mathbf{V}_p + \mathcal{D}_{7\times7}(\mathbf{V}_p)\right) \tag{12}$$

$$\mathbf{F}_o = \mathcal{P}\left(\mathcal{C}\left(\mathbf{F}_p,\ \mathbf{F}_s\right)\right). \tag{13}$$

Here, $\mathcal{P}$ represents the linear projection layer, implemented using a 1×1 convolution. The operator $\mathcal{C}$ refers to the concatenation along the channel dimension and $\mathcal{D}_{7\times7}$ denotes the depth-wise separable convolution with a kernel size of 7×7, which is used to introduce positional encoding into the Value matrix.

Therefore, by introducing the PFGM, the shared similarity weight between prior features and semantic features reduces the computational complexity. In addition, adopting the symmetric cross-attention mechanism to calculate respective attention weights allows the model to handle both prior and semantic features equally. The PFGM enables a more balanced and comprehensive integration of the two inputs with differences, thereby achieving feature complementarity.

C. Prior Edge Extraction Module

The SAM has learned extensive prior knowledge from large-scale datasets, enabling it to predict clear object boundaries. Notably, these segmentation results lack semantic information, making them unsuitable for directly implementing semantic segmentation tasks. To compensate for this issue, we present the PEEM method to extract edge information from the input images, which is then used to guide the network in obtaining clear and precise edges. The formula for edge extraction is as follows:

$$\mathbf{F}_e = \Psi(\mathbf{X}_s). \tag{14}$$

The symbol Ψ represents two BLC modules, each comprising batch normalization, a LeakyReLU activation, and a convolution layer with a kernel size of 1×1

$$\mathbf{E}_c = \mathcal{F}_{\text{Canny}}(\mathcal{F}_{\text{SAM}}(\mathbf{X}_{\text{in}})). \tag{15}$$

The symbol $\mathcal{F}_{\text{SAM}}$ denotes the lightweight SAM, which generates a result image where all objects are accurately segmented. Although this output contains clear boundary information, it lacks semantic content. $\mathcal{F}_{\text{Canny}}$ refers to the Canny edge detection operator, which extracts clear object edges $\mathbf{E}_c$ from the SAM output. This edge map $\mathbf{E}_c$ is employed to supervise the encoder's output $\mathbf{F}_e$, facilitating the generation of distinct object boundaries.

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D. Loss Function

The loss function of the PGNet consists of two components: supervised loss and unsupervised loss. In the final experiment, we assign the same weight of 0.5 to each part. During model training, all losses are computed using the cross-entropy loss, except for the edge loss in the unsupervised component. The formula for the cross-entropy loss is as follows:

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\log\left(\hat{y}_{i,c}\right) \tag{16}$$

where $i \in [1, 2, \ldots, N]$ represents the number of samples, $c \in [1, 2, \ldots, C]$ represents the number of categories, $y_{i,c}$ represents the true label, and $\hat{y}_{i,c}$ is the prediction results. The specific form of the loss function is as follows:

$$\mathcal{L} = \frac{1}{2}\mathcal{L}_s + \frac{1}{2}\mathcal{L}_u. \tag{17}$$

We follow the UniMatch [20] and most semi-supervised methods by setting the weights of the supervised and unsupervised losses to 0.5 each. The supervised loss $\mathcal{L}_s$ consists of the prediction loss and the auxiliary loss. The prediction loss is obtained by computing the cross-entropy loss between the model's final predictions and the GT labels.

The auxiliary loss is calculated using the output from the third layer of the backbone and the GT labels, with its weight controlled by the parameter $\lambda \in (0, 1)$. The formulas are as follows:

$$\mathcal{L}_s = \mathcal{L}_p + \lambda\mathcal{L}_{\mathrm{aux}} \tag{18}$$

where $\mathcal{L}_p$ is the prediction result loss and $\mathcal{L}_{\mathrm{aux}}$ is the auxiliary loss. We set $\lambda = 0.5$, and in subsequent ablation experiments, we will discuss the impact of the value of λ on the experimental results.

The unsupervised loss for the unlabeled data consists of two components. We apply perturbations of varying degrees to the unlabeled data, using the prediction results of weakly augmented data as pseudo-labels for the strongly augmented data, and compute the consistency loss between the two. Additionally, for strongly augmented unlabeled data, we introduce an edge supervision loss through the PEEM module to enhance the model's ability to perceive boundaries. The final unsupervised loss is composed of a linear combination of the consistency loss and the edge loss, as expressed below:

$$\mathcal{L}_u = \mathcal{L}_c + \mu\mathcal{L}_e. \tag{19}$$

$\mathcal{L}_c$ denotes the consistency loss and μ is used to control the weight of the edge supervision loss. The importance of edge supervision varies with the proportion of labeled data. Therefore, the value of μ is adjusted dynamically based on the labeling ratio. Experiments in the ablation study validate the correlation between μ and the labeling ratio. $\mathcal{L}_e$ represents the edge loss, which is calculated by the weighted binary cross entropy loss. The calculation formula is as follows:

$$\mathcal{L}_e = -\frac{1}{N}\sum_{i=1}^{N} w_i\left[y_i\log s_i + (1-y_i)\log(1-s_i)\right] \tag{20}$$

where $\mathcal{L}_e$ represents the edge supervision loss, N is the total number of samples, and w_i is the weight assigned to the ith sample, used to emphasize certain samples during training. $y_i \in \{0, 1\}$ denotes the GT label for the ith sample, and $s_i \in (0, 1)$ is the predicted probability for the ith sample after applying the sigmoid activation function. The term $y_i \log s_i$ calculates the contribution of positive samples ($y_i = 1$), while $(1 - y_i)\log(1 - s_i)$ accounts for the contribution of negative samples ($y_i = 0$). The overall loss $\mathcal{L}_e$ is averaged over all N samples.

IV. Experiment

A. Dataset

We validate the effectiveness of our proposed method using three widely recognized remote sensing datasets: the ISPRS Vaihingen dataset, the ISPRS Potsdam dataset, and the LoveDA dataset.

1) ISPRS Vaihingen: The ISPRS Vaihingen dataset comprises 33 high-resolution orthophotos. The size of the images in the dataset ranges from 1996 × 1995 pixels to 3816 × 2550 pixels, with a ground resolution of approximately 9 cm per pixel. Each orthophoto includes three channels: red, green, and near-infrared, each providing rich spectral information. Additionally, the dataset contains near-infrared images, digital surface model (DSM) data, and normalized DSM (nDSM) data. The dataset is annotated with six classes, comprising five foreground classes and one background class. In the set of 33 images, 16 are designated for training, while the remaining 17 are used for testing. In our experiments, the entire training set is further divided into two groups for evaluation. Specifically, 1/16, 1/8, and 1/4 of the training set are randomly selected as labeled data, with the remaining images used as unlabeled data.

2) ISPRS Potsdam: The ISPRS Potsdam dataset consists of 38 images with corresponding labels. The dataset includes RGB images containing three spectral channels: red, green, and blue. Each image has a size of 6000 × 6000 pixels, with a ground resolution of approximately 5 cm per pixel. The dataset is annotated with six classes, comprising five foreground classes and one background class. In our experiments, due to incorrect annotations in image 7_10, we used 23 images as the training set and 14 images as the test set. We randomly select annotation ratios of 1/12, 1/8, and 1/4 from the training set for experiments and perform quantitative comparisons with previous studies.

3) LoveDA Dataset: The dataset is a land-cover dataset for urban and rural areas in China, comprising 5987 high-resolution remote sensing images. Each image has a resolution of 1024 × 1024 pixels, with each pixel corresponding to an actual distance of 0.3 m. The dataset includes eight categories: background, building, road, water, barren, forest, agriculture, and no data. The no-data class is treated as an ignored class. Since the test set in this dataset does not include corresponding labels, we use the validation set as the test set. Following previous work [78], we randomly select 5%, 10%, and 20% of the training set as labeled data, with the remaining images used as unlabeled data.

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TABLE I

COMPARED WITH SEMI-SUPERVISED METHODS ON THE ISPRS VAIHINGEN DATASET. THE BEST PERFORMANCE IS IN BOLD. THE SYMBOL "-" INDICATES THAT THE ORIGINAL PAPER DID NOT PROVIDE DATA AND DID NOT PUBLICLY RELEASE THE CODE

Annotation Ratio	Method	F1 score per category(%)					Overall Metrics		
		Imp.surf.	Building	Low veg.	Tree	Car	mF1(%)	OA(%)	mIoU(%)
1/16	Sup only	82.59	83.02	65.58	79.38	61.81	74.48	77.83	60.29
	MT [68]	85.82	88.16	68.95	80.37	64.98	77.66	81.08	64.43
	Fixmatch [18]	86.55	88.73	69.05	80.69	65.72	78.15	82.34	65.27
	ST [22]	85.76	87.94	66.92	79.44	63.79	76.10	80.41	63.33
	UNetFormer [69]	84.32	88.03	65.31	77.36	63.32	75.66	79.31	64.21
	ClassMix [70]	86.60	89.26	68.92	78.77	69.11	78.53	80.94	65.50
	AEL [71]	87.52	90.73	68.92	78.75	73.54	79.89	81.79	67.33
	U2PL [72]	87.70	91.34	67.09	77.90	72.66	79.34	81.76	66.84
	LPCR [59]	87.19	90.31	71.00	81.14	75.29	80.99	82.73	68.71
	ECAE [73]	85.51	90.19	72.77	84.11	74.52	81.42	83.35	69.20
	UniMatch [20]	89.28	93.11	77.07	86.42	77.79	84.73	87.10	74.03
	MDMCR [60]	88.77	93.56	75.31	84.03	74.30	83.19	86.70	71.52
	PGNet (Ours)	**90.43**	**94.08**	**78.52**	**86.77**	**81.69**	**86.30**	**88.09**	**76.33**
1/8	Sup only	84.93	87.93	73.93	82.07	68.43	79.46	82.04	66.61
	MT [68]	87.00	90.58	75.75	82.62	74.99	82.19	83.94	70.26
	Fixmatch [18]	87.32	90.69	76.52	82.86	76.33	82.74	84.67	71.33
	ST [22]	86.83	90.92	74.85	81.87	75.62	82.02	83.55	70.03
	UNetFormer [69]	85.82	89.34	74.17	82.09	74.17	81.11	80.79	70.16
	ClassMix [70]	86.63	90.55	76.66	83.74	76.77	83.07	84.43	71.44
	AEL [71]	87.78	92.25	76.12	82.34	77.11	83.69	84.55	71.64
	U2PL [72]	87.59	91.22	77.96	81.24	79.58	83.18	85.43	72.34
	LPCR [59]	88.30	92.19	78.57	82.87	79.29	84.22	85.45	73.14
	ECAE [73]	86.50	91.22	72.94	84.30	76.48	82.29	84.04	70.45
	UniMatch [20]	89.63	93.22	77.87	86.52	76.58	84.76	87.40	74.11
	MDMCR [60]	89.40	93.91	**79.42**	86.51	78.74	85.59	87.27	74.48
	HARC [74]	86.67	90.83	74.53	83.58	71.31	81.38	-	69.26
	PGNet (Ours)	**90.61**	**94.24**	79.26	**86.87**	**81.85**	**86.57**	**88.36**	**76.73**
1/4	Sup only	87.69	91.09	76.90	84.23	76.71	83.33	85.01	71.85
	MT [68]	88.68	91.96	76.62	84.93	79.22	84.69	86.18	73.80
	Fixmatch [18]	88.33	92.52	78.93	84.96	79.96	84.94	86.37	74.32
	ST [22]	88.74	92.74	79.17	84.92	80.22	85.16	86.39	74.50
	UNetFormer [69]	89.07	91.94	78.61	84.69	80.06	84.87	85.81	74.64
	ClassMix [70]	89.16	92.47	79.54	85.54	80.24	85.39	86.75	74.54
	AEL [71]	88.93	92.74	80.14	85.15	80.33	85.46	86.70	74.93
	U2PL [72]	89.30	93.32	80.76	85.52	81.45	86.07	87.19	75.86
	LPCR [59]	89.57	93.47	81.01	85.93	81.94	86.39	87.48	76.34
	ECAE [73]	86.42	91.64	74.89	84.99	76.95	82.98	84.64	71.39
	UniMatch [20]	89.31	93.26	77.16	86.49	77.10	85.47	87.52	74.96
	MDMCR [60]	89.66	94.12	**81.67**	86.17	81.56	86.63	87.62	77.03
	HARC [74]	88.08	91.54	78.20	84.55	75.82	83.64	-	72.32
	PGNet (Ours)	**90.62**	**94.35**	79.09	**86.86**	**83.39**	**86.86**	**88.34**	**77.17**

B. Evaluation Metrics and Experimental Settings

1) Evaluation Metrics: We use three metrics to evaluate the performance of our model on the ISPRS Vaihingen and ISPRS Potsdam datasets: overall accuracy (OA), mean intersection over union (mIoU), and $F1$ score. For the LoveDA dataset, the evaluation metrics include $F1$ score, mIoU, and the kappa coefficient. The formulas for OA, mIoU, and $F1$ score are as follows:

$$\mathrm{OA} = \frac{\sum_{k=1}^{N} \mathrm{TP}_k}{\sum_{k=1}^{N} (\mathrm{TP}_k + \mathrm{FP}_k + \mathrm{TN}_k + \mathrm{FN}_k)} \tag{21}$$

$$\mathrm{mIoU} = \frac{1}{N} \sum_{k=1}^{N} \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FP}_k + \mathrm{FN}_k} \tag{22}$$

$$F1_k = 2 \times \frac{\mathrm{precision}_k \times \mathrm{recall}_k}{\mathrm{precision}_k + \mathrm{recall}_k} \tag{23}$$

$$\mathrm{precision}_k = \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FP}_k} \tag{24}$$

$$\mathrm{recall}_k = \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FN}_k}. \tag{25}$$

In these formulas, TP_k (true positives) represents the number of correctly classified pixels for class k; FP_k (false positives) is the number of pixels incorrectly classified as class k; FN_k (false negatives) is the number of pixels of class k that were not classified correctly; and TN_k (true negatives) represents all other correctly classified pixels not belonging to class k. N is the total number of classes, excluding the background class. The $F1$ score is calculated for each class and averaged to evaluate the overall performance.

For the LoveDA dataset, we additionally use the kappa coefficient, which accounts for the possibility of chance agreement. The kappa coefficient and its related term PRE are defined as follows:

$$\mathrm{PRE} = \frac{(\mathrm{TP}_k + \mathrm{FN}_k)(\mathrm{TP}_k + \mathrm{FP}_k) + (\mathrm{TN}_k + \mathrm{FN}_k)(\mathrm{TN}_k + \mathrm{FP}_k)}{(\mathrm{TP}_k + \mathrm{TN}_k + \mathrm{FP}_k + \mathrm{FN}_k)^2} \tag{26}$$

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TABLE II

COMPARED WITH SEMI-SUPERVISED METHODS ON THE ISPRS POTSDAM DATASET. THE BEST PERFORMANCE IS IN BOLD. THE SYMBOL "-" INDICATES THAT THE ORIGINAL PAPER DID NOT PROVIDE DATA AND DID NOT PUBLICLY RELEASE THE CODE

Annotation Ratio	Method	F1 score per category(%)					Overall Metrics		
		Imp.surf.	Building	Low veg.	Tree	Car	mF1(%)	OA(%)	mIoU(%)
1/12	Sup only	84.47	90.29	76.43	76.79	87.05	83.01	89.06	71.38
	MT [68]	85.96	91.19	76.96	78.99	87.33	84.09	82.27	72.95
	Fixmatch [18]	86.42	90.55	78.65	79.62	87.82	84.33	82.62	73.15
	ST [22]	86.49	92.14	77.88	78.13	86.76	84.28	82.53	73.24
	UNetFormer [69]	84.57	91.00	77.09	79.33	87.63	84.60	86.13	72.96
	ClassMix [70]	86.73	92.49	78.77	80.44	87.91	85.27	83.34	74.68
	AEL [71]	87.98	92.98	79.34	80.70	88.63	85.92	84.34	75.68
	U2PL [72]	87.52	92.89	78.85	80.36	88.25	85.57	83.87	75.17
	LPCR [59]	88.02	93.46	79.13	81.58	88.53	86.14	84.36	76.03
	ECAE [73]	90.13	95.10	77.98	81.05	84.92	85.04	88.01	75.57
	UniMatch [20]	87.39	92.04	87.55	80.61	82.32	85.98	85.70	76.81
	MDMCR [60]	87.04	94.77	83.46	**84.10**	**88.76**	87.62	85.44	77.21
	PGNet (Ours)	**92.42**	**96.13**	**93.89**	82.47	88.42	**90.67**	**92.31**	**83.27**
1/8	Sup only	86.81	91.37	78.04	78.92	87.44	84.51	82.94	73.56
	MT [68]	88.03	92.84	80.01	80.92	88.63	86.09	84.49	75.92
	Fixmatch [18]	87.91	93.60	79.94	80.41	88.21	85.67	83.39	75.84
	ST [22]	87.50	92.45	79.79	81.42	88.49	85.93	84.39	75.63
	UNetFormer [69]	88.60	92.73	78.29	81.63	88.37	85.92	84.72	75.97
	ClassMix [70]	88.33	92.88	80.00	80.71	89.13	86.21	84.59	76.11
	AEL [71]	87.51	92.84	80.18	82.06	88.94	86.30	84.76	76.20
	U2PL [72]	88.16	92.99	80.71	82.10	89.21	86.63	85.16	76.71
	LPCR [59]	88.65	93.31	81.30	82.59	89.49	87.07	85.57	77.33
	ECAE [73]	**91.33**	95.93	79.76	82.00	87.41	87.29	89.51	77.94
	UniMatch [20]	86.53	93.82	86.01	81.09	86.43	87.18	85.84	78.24
	MDMCR [60]	88.28	94.95	83.15	**84.91**	88.70	87.99	85.82	78.80
	HARC [74]	89.27	93.05	81.74	83.55	**90.71**	87.66	-	78.30
	FusionMatch [75]	-	-	-	-	-	88.37	87.36	79.37
	PGNet (Ours)	91.15	**96.81**	**93.72**	83.53	88.88	**90.82**	**92.56**	**83.49**
1/4	Sup only	87.86	91.95	79.45	81.77	88.98	86.00	84.30	75.76
	MT [68]	88.93	93.86	80.79	81.59	89.71	86.97	85.44	77.30
	Fixmatch [18]	89.27	93.06	80.92	81.97	89.67	86.98	85.60	77.42
	ST [22]	89.35	93.92	80.98	81.82	89.52	87.12	85.65	77.53
	UNetFormer [69]	88.96	93.76	80.42	81.37	89.32	86.76	85.97	77.79
	ClassMix [70]	89.40	94.21	81.49	82.74	89.72	87.51	86.04	78.13
	AEL [71]	89.63	94.12	82.27	83.39	89.47	87.78	86.34	78.49
	U2PL [72]	89.81	94.27	82.07	83.53	89.90	87.92	86.55	78.73
	LPCR [59]	90.00	94.82	83.19	84.24	90.26	88.50	87.21	79.64
	ECAE [73]	91.58	96.32	81.93	82.53	88.39	88.15	89.84	78.68
	UniMatch [20]	90.15	94.47	87.73	80.72	87.08	88.07	86.36	78.98
	MDMCR [60]	90.61	95.15	84.50	**86.16**	90.97	89.47	88.03	81.01
	HARC [74]	89.41	93.23	82.95	85.26	**91.16**	88.40	-	79.42
	FusionMatch [75]	-	-	-	-	-	89.22	87.83	80.50
	PGNet (Ours)	**93.12**	**96.44**	**94.57**	81.58	90.21	**91.18**	**93.21**	**84.20**

$$\text{kappa} = \frac{\text{OA} - \text{PRE}}{1 - \text{PRE}}. \tag{27}$$

For the ISPRS Vaihingen and Potsdam datasets, we ignore the background class in the calculations of OA, mIoU, and $F1$ score, consistent with previous studies [59], [60]. For the LoveDA dataset, background class pixels are included in all metric calculations.

2) Experimental Settings: Our experiments were conducted on an NVIDIA A40 GPU with 40 GB of memory. The learning rate was set to 0.0001, and ResNet-101 was used as the backbone of the model. The AdamW optimizer was employed for training. To facilitate processing, the Vaihingen and Potsdam datasets were randomly cropped to a size of 512 × 512, while the LoveDA dataset was kept at its original size of 1024 × 1024. Data augmentation techniques were applied to all datasets, and the augmentation method is consistent with the baseline UniMatch. The model was trained for a total of 150 epochs, with a confidence threshold set to 0.95. The lightweight SAM uses the EfficientSAM. We adopted UniMatch as the baseline framework, with DeepLabV3+ as the semantic segmentation model.

C. Experimental Results Analysis

In this section, we provide a detailed analysis of our model's performance on three remote sensing datasets. Through systematic quantitative comparisons, we evaluate the model's performance across various key metrics. Additionally, qualitative analyses are conducted to further verify the model's

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TABLE III
COMPARISON OF VARIOUS METHODS ON THE LOVEDA DATASET. THE BEST PERFORMANCE IS IN BOLD

Annotation Ratio	Method	F1 score per category(%)							Overall Metrics		
		Background	Building	Road	Water	Barren	Forest	Agriculture	mIoU(%)	mF1(%)	Kappa(%)
5%	CCT [76]	63.76	61.70	63.76	75.74	39.79	45.00	61.40	42.48	58.74	48.50
	CPS [77]	65.68	66.35	64.84	75.21	8.93	53.04	64.24	42.12	56.90	49.76
	SegMind [78]	66.07	63.79	57.98	77.41	43.22	40.98	67.18	43.31	59.52	52.36
	U2PL [72]	68.92	69.39	67.52	79.34	28.30	**55.24**	64.77	46.42	61.93	53.91
	LSST [79]	67.97	62.70	68.99	80.69	**50.21**	52.72	65.41	47.91	64.10	54.34
	FixMatch [18]	67.70	67.89	67.23	76.98	45.29	51.47	71.37	47.88	63.99	57.35
	UniMatch [20]	69.40	74.13	**71.18**	80.15	38.96	52.79	73.19	50.27	65.68	59.52
	DWL [62]	65.55	70.97	68.01	82.00	45.51	53.57	68.52	48.99	64.88	55.97
	PGNet (Ours)	**70.12**	**74.23**	69.82	**82.28**	41.86	49.78	**76.22**	**51.10**	**66.33**	**61.62**
10%	CCT [76]	61.18	62.28	64.54	72.71	39.24	49.05	62.13	42.29	58.73	47.62
	CPS [77]	67.81	70.91	68.92	69.60	31.07	54.64	69.49	45.91	61.78	54.79
	SegMind [78]	68.29	69.62	57.30	81.01	43.85	48.42	68.92	46.47	62.49	55.78
	U2PL [72]	67.94	69.66	69.65	72.46	45.49	56.68	68.33	48.00	64.32	55.37
	LSST [79]	67.28	66.22	68.96	82.25	42.83	53.21	68.47	48.32	64.17	55.65
	FixMatch [18]	66.99	74.13	69.41	79.40	40.34	52.74	72.93	49.53	65.13	58.62
	UniMatch [20]	67.16	73.54	**70.17**	81.26	42.07	56.28	72.21	50.49	66.10	58.72
	DWL [62]	66.61	72.33	70.04	82.59	**46.89**	**58.64**	69.39	50.87	66.64	57.53
	PGNet (Ours)	**69.43**	**75.98**	69.84	**83.44**	42.30	54.87	**74.55**	**51.96**	**67.20**	**61.70**
20%	CCT [76]	66.88	51.57	60.39	77.12	32.16	48.42	58.76	41.90	56.44	51.61
	CPS [77]	68.93	69.42	69.63	70.56	32.13	56.48	70.24	46.02	62.49	55.28
	SegMind [78]	69.32	62.46	56.33	**82.65**	44.90	57.03	74.06	47.15	63.25	56.93
	U2PL [72]	67.91	73.79	66.84	81.79	41.52	52.39	68.63	49.03	64.70	56.33
	LSST [79]	67.10	67.24	53.27	79.07	36.37	50.70	61.62	43.37	59.34	53.66
	FixMatch [18]	68.13	67.49	67.75	80.48	47.08	57.09	69.78	49.35	65.39	57.56
	UniMatch [20]	65.94	73.68	**69.70**	79.87	49.52	**57.15**	**75.31**	51.54	67.31	60.08
	DWL [62]	67.20	75.73	68.81	80.86	**49.78**	55.52	72.29	51.43	67.17	58.07
	PGNet (Ours)	**70.91**	**76.87**	69.50	82.60	46.39	56.12	74.85	**52.85**	**68.18**	**62.19**

effectiveness in specific image segmentation tasks. These comprehensive evaluation results strongly highlight the outstanding performance of our model, even under scenarios with a limited number of labeled samples.

1) Comparison on the Vaihingen Dataset: Table I presents the performance of our model on various metrics for the Vaihingen dataset. As shown in the table, our model achieves superior results across key indicators compared to the previous state-of-the-art (SOTA) methods. With annotation ratios of 1/4, 1/8, and 1/16, our model achieves the best performance across all overall metrics. In terms of mIoU, it improves by 0.14%, 2.25%, and 2.30%, respectively, compared to the second-best model. We observed that as the amount of labeled data decreases, the advantages of our model become more pronounced. However, in the comparison of individual indicators, the performance of the low vegetation class is less competitive at the $1/8$ and $1/4$ labeled data ratios. We attribute this to the unclear boundaries between low vegetation and trees in the Vaihingen dataset, which may introduce interference during classification.

As shown in Fig. 4, in addition to the quantitative comparison, we conducted a visual analysis on the $1/16$ annotation ratio of the Vaihingen dataset. Challenging regions where our method demonstrates significant improvements are highlighted with red boxes. Compared with previous representative semi-supervised methods, the segmentation results of our model are noticeably closer to the GT labels.

2) Comparison on the Potsdam Dataset: As shown in Table II, our model achieves superior performance on most metrics for the Potsdam dataset. Notably, when labeled data is scarce, our model exhibits more significant performance improvements compared to previous methods, highlighting its effectiveness in low-label scenarios. At labeled data ratios of $1/4$, $1/8$, and $1/12$, our model outperforms the previous best methods in the key metrics of m$F1$, OA, and mIoU, with mIoU improvements of 3.19%, 4.12%, and 6.06%, respectively.

For the $F1$ score of individual classes, our model is slightly inferior to the previous SOTA method for the tree class. We attribute this to the lack of clear boundaries between trees and the background, caused by the presence of numerous branches and sparse foliage. To intuitively demonstrate the performance of our model, we conducted a visual comparison on the Potsdam dataset at a $1/16$ annotation scale, as shown in Fig. 5. The visualization results show that the predictions of our model are closer to the GT, particularly in areas with complex categories, such as those highlighted in the red boxes. This demonstrates our model's ability to capture details, handle complex scenes, and recover finer structures.

As illustrated in Fig. 6, we extracted the features from the last layer of the backbone network for different methods and performed t-SNE dimensionality reduction on these features using the $1/16$ and $1/12$ annotation ratio weights of the Vaihingen and Potsdam datasets. The visualization

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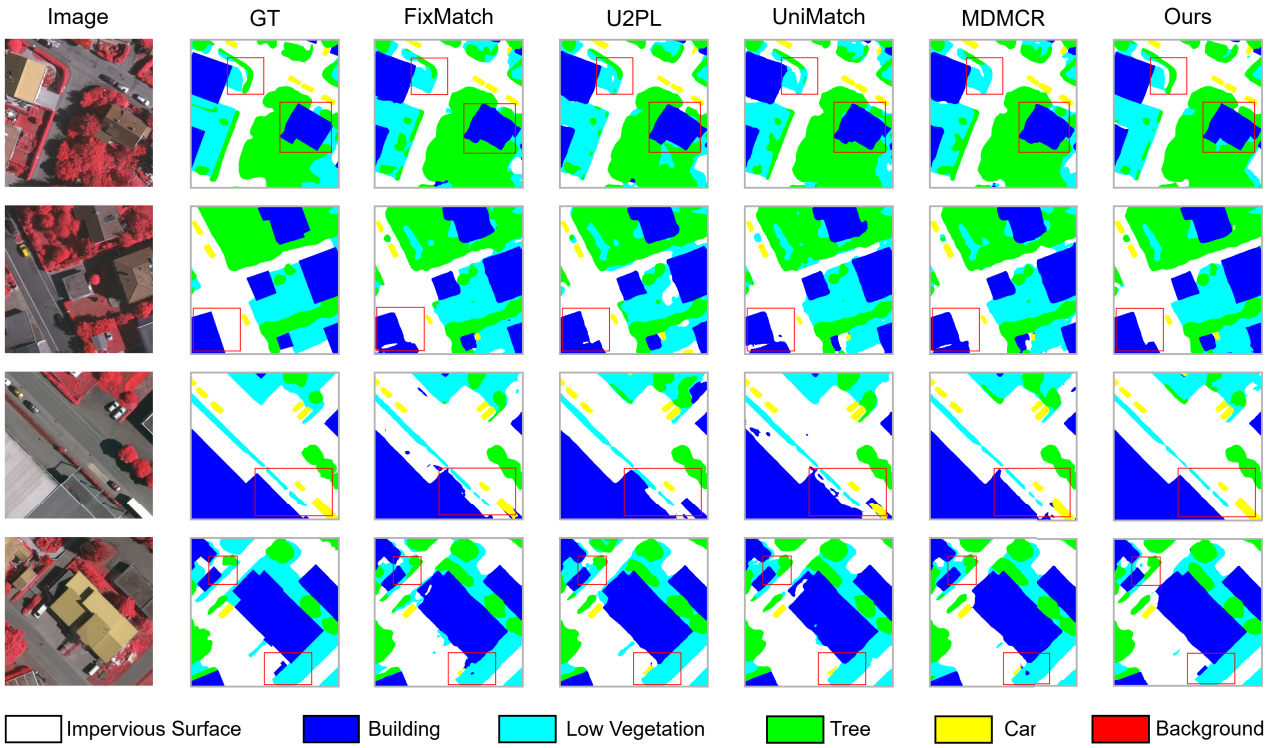


Fig. 4. Visual comparison on the 1/16 labeled Vaihingen dataset. The red boxes highlight the areas where our method achieves significant improvements.

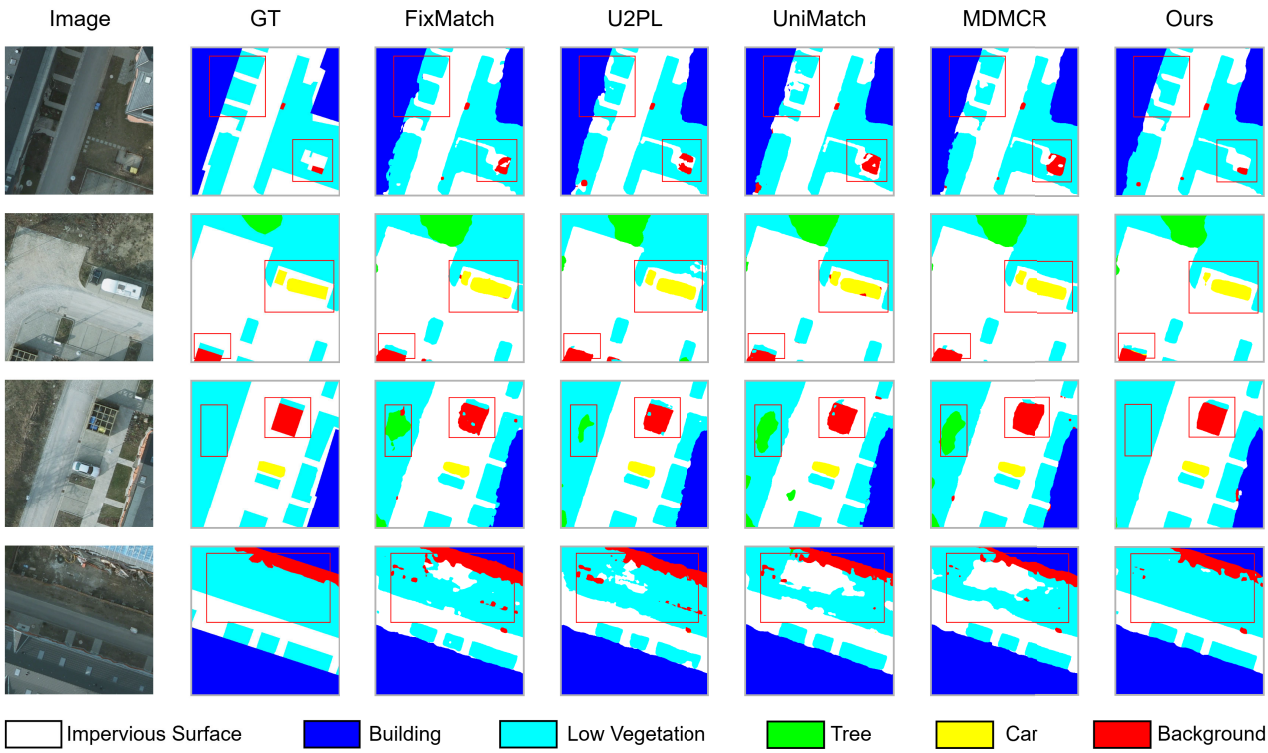


Fig. 5. Visual comparison on the 1/12 labeled Potsdam dataset. The red boxes highlight the areas where our method achieves significant improvements.

results reveal that our method demonstrates a stronger ability to focus on specific categories, further validating its effectiveness.

3) Comparison on the LoveDA Dataset: The segmentation performance of our method on the LoveDA dataset, compared to SOTA methods, is summarized in Table III. The results

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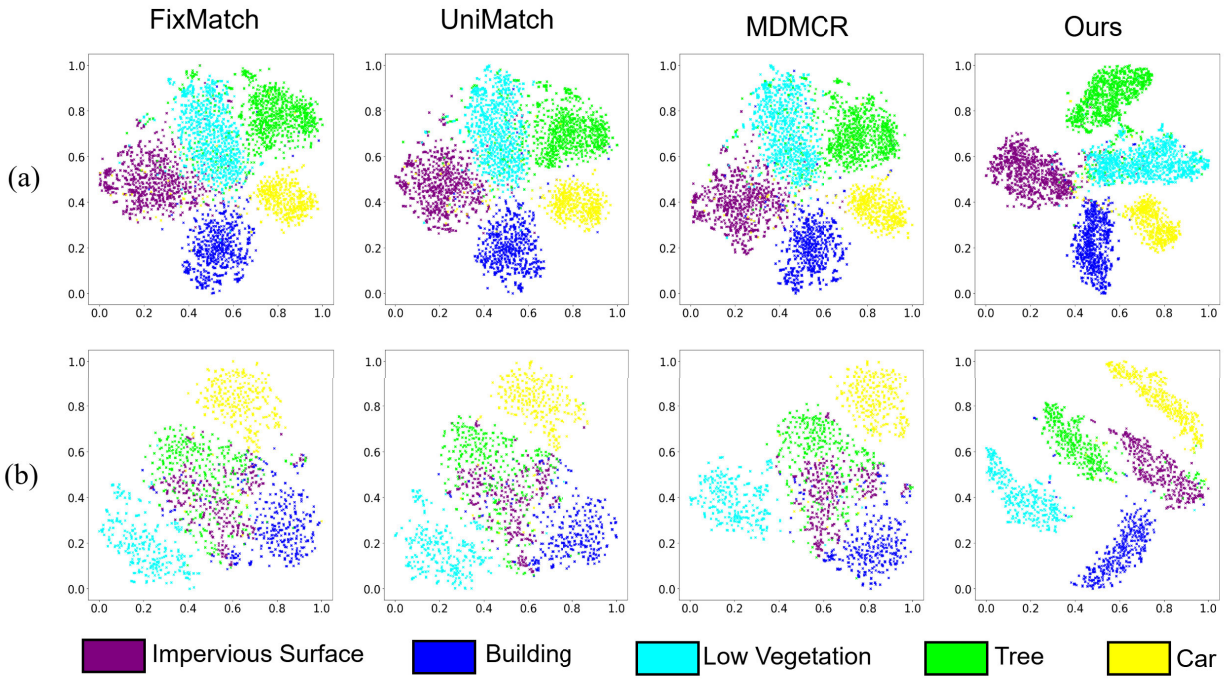


Fig. 6. (a) First row represents the comparison of different methods on t-SNE visualization with a 1/12 annotation ratio for the Potsdam dataset. (b) Second row represents the comparison of different methods on t-SNE visualization with a 1/16 annotation ratio for the Vaihingen dataset.

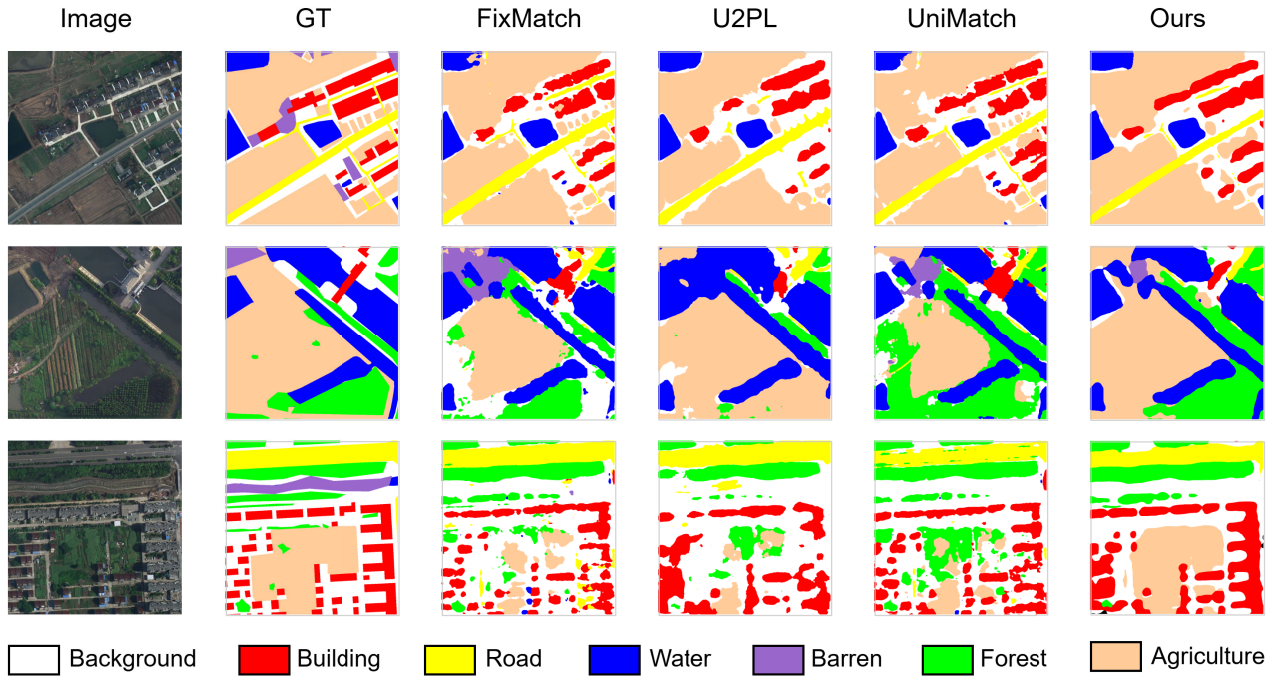


Fig. 7. Visual comparison with other methods on the 5% annotation ratio of the LoveDA dataset. It can be seen that our method significantly improves the segmentation results.

indicate that our approach achieves superior performance across all three key composite metrics. These findings underscore the importance of integrating prior edge information and prior feature diagnosis into our model.

A visual comparison of the segmentation results on 5% of the labeled LoveDA dataset is shown in Fig. 7. It can be observed that our method captures and processes finer details than previous approaches, further validating the critical role of prior information in enhancing semi-supervised semantic segmentation.

D. Ablation Experiment

The primary advantages of our PGNet stem from the PFGM and PEEM modules. PFGM extracts prior features

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TABLE IV
PERFORMANCE GAINS FROM DIFFERENT MODULES AND THEIR COMBINATIONS

Method	mF1 (%)	OA (%)	mIoU (%)
EfficientSAM	-	-	81.70
Baseline	87.84	87.70	78.61
Baseline + PEEM	89.35	91.27	81.10
Baseline + PFGM	90.19	92.21	82.51
Baseline + PEEM + PFGM	**91.82**	**93.63**	**83.27**

TABLE V
ABLATION STUDY OF FEATURE FUSION METHODS IN THE PFGM

Method	mF1 (%)	OA (%)	mIoU (%)
Add	89.64	90.93	81.66
Concat	89.39	91.07	81.20
Hadamard product	89.52	90.69	81.46
Cross-Attention	89.55	90.79	81.47
Symmetric Cross-Attention	**89.85**	**91.92**	**81.94**

TABLE VI
PERFORMANCE COMPARISON OF DIFFERENT μ VALUES AT DIFFERENT ANNOTATION RATIO ON THE POTSDAM DATASET

Annotation Ratio	μ	mF1(%)	OA(%)	mIoU(%)
1/12	6	88.73	89.41	80.07
	12	89.21	90.43	80.86
	24	**89.35**	**91.27**	**81.10**
	48	89.24	90.43	80.92
1/8	4	89.53	90.82	81.36
	8	89.11	89.28	80.63
	16	**89.98**	**92.10**	**82.16**
	32	89.59	90.93	81.48
1/4	1	90.19	91.79	82.48
	4	**90.93**	**92.28**	**83.70**
	8	90.23	91.01	82.56
	16	90.14	91.05	82.45

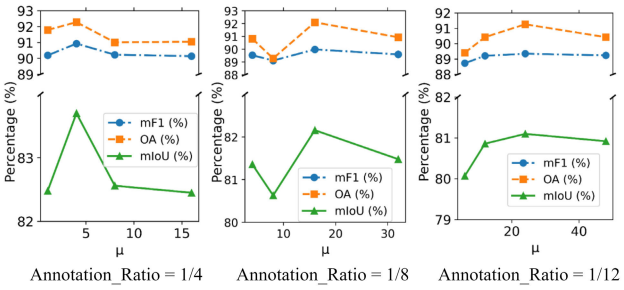


Fig. 8. Ablation analysis of the edge loss weight μ under different annotation ratios on the Potsdam dataset.

to guide model training, while PEEM captures edge priors from unlabeled data, enhancing the model's ability to perceive boundaries in such data. Consequently, we focused on designing the model's loss function to maximize these benefits. In this section, we validate the impact of each module on the overall performance through a series of ablation experiments on the Potsdam dataset. Furthermore, we examine the parameters involved in the loss function to identify the optimal configuration that yields the maximum performance gain. The ablation experiment section provide a detailed introduction and analysis of each module and its corresponding experimental results to verify their effectiveness and necessity.

1) Ablation Experiment on Each Module of the PGNet: To demonstrate the effectiveness of the PGNet framework, we embedded the PFGM and PEEM modules into the baseline model, which uses ResNet-101 as its backbone. Experimental results indicate that on the Potsdam dataset, incorporating these two modules leads to a significant improvement in the mIoU metric compared to the original baseline model. In addition, we compared mIoU using the calculation method officially provided by EfficientSAM.

As shown in Table IV, taking the 1/12 annotation ratio of the Potsdam dataset as an example, adding each of the proposed modules individually results in performance improvements. Furthermore, combining both modules yields significant enhancements in key metrics, demonstrating their synergy. Note that compared to the original EfficientSAM, we have improved it by 1.57%.

2) Ablation Study of Feature Fusion Methods in the PFGM: We conducted a comprehensive ablation study on the feature fusion methods in the PFGM. We compared our proposed symmetric cross-attention mechanism with other fusion techniques, including addition, concatenation, Hadamard product, and standard cross-attention. The experimental results, shown in Table V, indicate that our symmetric cross-attention method achieves the best segmentation performance. This advantage arises from its ability to project and compute attention weights between the input prior and semantic features, allowing for more effective utilization of these complementary features.

3) Ablation Experiment on Edge Loss Weight μ: In (19), the parameter μ controls the weight of the boundary loss for unlabeled data. Unlabeled data is optimized using both the consistency loss and the boundary loss, while labeled data is supervised by the GT during training. Considering the influence of different annotation ratios on model training, we hypothesize that dynamically adjusting the edge loss weight is essential, as it directly affects the balance between the loss contributions from labeled and unlabeled data. To evaluate this hypothesis, we conducted extensive experiments with various labeled data ratios on the Potsdam dataset, analyzing the impact of different values of μ on model performance.

As shown in Table VI and Fig. 8, the value of μ exhibits a clear pattern: as the labeled data ratio increases, a smaller μ value results in better performance. Upon analysis, we attribute this behavior to the fact that when the amount of labeled data increases, the model benefits more from direct supervision by the GT. Consequently, the boundary loss within the unsupervised loss should be assigned a smaller weight to maintain an appropriate balance in the loss function.

4) Ablation Study of λ: The parameter λ represents the weight of the auxiliary loss for labeled data in (18). When calculating the loss for labeled data, relying solely on the output from the final layer of the model to compute the loss with the GT may result in slow convergence. To resolve this, we incorporate the output from the final layer of the

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TABLE VII
ABLATION EXPERIMENT OF λ VALUE ON THE 1/12 ANNOTATION RATIO OF THE POTSDAM DATASET

λ	mF1 (%)	OA (%)	mIoU (%)
0.1	88.37	87.46	79.43
0.3	88.75	89.80	80.11
0.5	**89.10**	**90.23**	**80.65**
0.7	88.71	88.51	80.01

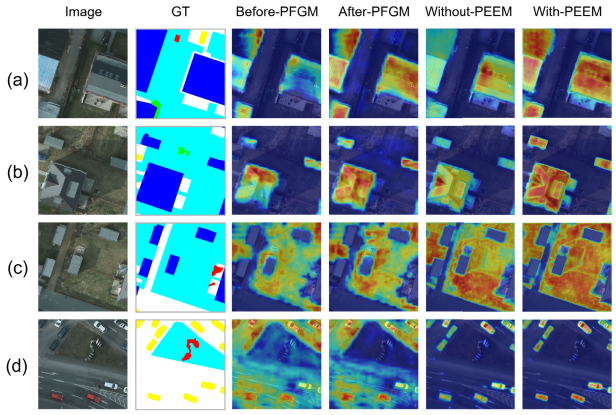


Fig. 9. Ablation experiments were conducted on the PEEM and PFGM. We used the Grad-CAM visualization method to visualize the feature maps before and after the PFGM, as well as the same layer feature maps with and without the PEEM. (a) and (b) Grad-CAM for the building class. (c) and (d) Grad-CAM for the low vegetation class and the car class, respectively.

TABLE VIII
IMPACT OF THE LIGHTWEIGHT SAM ENCODER ON PERFORMANCE

Method	mF1 (%)	OA (%)	mIoU (%)
Baseline	87.84	87.70	78.61
Baseline + PFGM(w/o. SAM)	89.41	91.32	81.22
Baseline + PFGM(Ours)	**90.19**	**92.21**	**82.51**
Baseline + PEEM(w/o. SAM)	88.45	91.08	79.63
Baseline + PEEM(Ours)	89.35	91.27	81.10

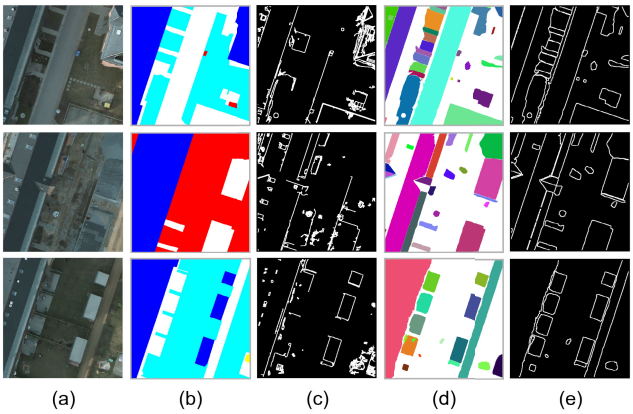


Fig. 10. Visual comparison of the edges extracted directly from the original image and those extracted from the efficient SAM segmentation results. (a) Input image. (b) GT. (c) Edge map directly extracted from the image using the Canny operator. (d) and (e) Segmentation result of the efficient SAM and the edges extracted from the segmentation result, respectively.

backbone network for joint supervision. Introducing additional supervision signals enables the model to learn more informative features during training, thereby improving its overall performance. As shown in Table VII, we find that the optimal value for λ is 0.5. Therefore, λ is set to 0.5 in all subsequent experiments.

5) Discuss the Advantages of Combining SAM With the PEEM and PFGM: To intuitively demonstrate the advantages brought by the combination of the SAM with the PEEM and PFGM, we conducted the following ablation experiments. For the PFGM, we visualized the feature maps before and after applying it. For the PEEM, we compared the visualizations of the same feature maps with and without the PEEM. The results are shown in Fig. 9. The visualization results clearly show that using the PEEM and PFGM significantly enhances the network's feature responses to various object categories. The focused regions in the feature maps become clearer, and the details are more refined.

6) Ablation Experiment of the SAM: To demonstrate the feasibility of incorporating prior information through the SAM, we conducted additional ablation experiments on these two modules. For the PEEM, we directly applied the Canny operator to extract the edges of the input image, providing edge supervision to the network's segmentation results. For the PFGM, we removed the feature input extracted by the lightweight SAM encoder, such that both inputs to the PFGM are the output features from the ResNet-101 encoder. The experimental results as shown in Table VIII, the "w/o." represents "without." The results indicate that the feature priors obtained from the SAM encoder significantly improve the model's semantic segmentation performance.

In addition, we conducted additional qualitative analysis for the PEEM module. We visualized the edges extracted directly using the Canny operator from the input image and those obtained from the lightweight SAM segmentation results. As shown in Fig. 10, the segmentation results from the lightweight SAM produce clearer and more accurate object edges, providing better guidance for the segmentation network to accurately identify object boundaries.

Additionally, to examine the impact of different SAM versions on prior information extraction, we applied various versions of the SAM to process the PEEM module. The experimental results are presented in Table IX. From the results, it is evident that among all lightweight SAM versions, EfficientSAM shows the most significant advantage in extracting prior information.

TABLE IX
IMPACT OF DIFFERENT VERSIONS OF SAM ON MODEL PERFORMANCE

Method	mF1 (%)	OA (%)	mIoU (%)
FastSAM	87.75	88.93	78.54
MobileSAM	90.38	92.33	82.86
EfficientSAM	**91.82**	**93.63**	**83.27**

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TABLE X
COMPARISON WITH OTHER METHODS IN TERMS OF PARAMETERS, COMPUTATIONAL COST (GFLOPS), AND INFERENCE TIME IS PRESENTED

Method	GFLOPs	Params(M)	Inference time(s)
ST++	732.6	**40.4**	0.036
FixMatch	384.4	59.5	**0.034**
U2PL	570.4	66.8	0.049
UniMatch	384.4	59.5	**0.034**
MDMCR	385.6	59.5	0.039
Ours	**372.3**	68.9	0.056

E. Efficiency Comparison

We compare the proposed model with other models in terms of parameter count, computational cost (GFLOPs), and inference time. To ensure the integrity and fairness of the experiment, all models use a unified input specification: the pixel size is 1024×1024, and the model processes one image at a time. It is worth noting that for the two comparison models Fixmatch and UniMatch, their network architectures are consistent, and the only difference is the different perturbation methods used, so they have the same number of parameters and computation.

As shown in Table X, the model parameters of the PGNet are comparable to other SOTA methods. Although the PGNet does not have an advantage in inference time, it achieves the lowest computational cost while delivering the best performance. Specifically, we reduce the computational cost of the model by applying downsampling operations to the feature layers in the network backbone and performing feature interactions at lower resolutions, ultimately achieving the lowest computational cost of 372.3 GFLOPs.

V. CONCLUSION

In this article, we propose PGNet, a semi-supervised semantic segmentation network specifically designed for remote sensing images. The PFGM in the PGNet effectively aggregates reliable prior information from the image into both shallow and deep features. In addition, our PEEM extracts reliable edge priors to enhance the model's boundary perception ability for unlabeled data. By dynamically adjusting the boundary supervision weights, we identify the optimal parameter configuration for achieving the best performance. Our model achieves excellent results on the ISPRS Vaihingen, ISPRS Potsdam, and LoveDA datasets. Even with minimal labeled data, the mIoU of our model increases by 2.30%, 6.06%, and 0.83% on these datasets, respectively. Our PGNet is specifically designed for semi-supervised remote sensing image segmentation scenarios with limited labeled data. However, its effectiveness in unsupervised tasks without labeled data or in weakly supervised tasks that rely solely on image-level labels or bounding box annotations still requires further validation. In future work, our goal is to further explore how to effectively apply prior knowledge from large models to more challenging remote sensing semantic segmentation tasks and experimentally validate these approaches. We hope that this research will not only promote the development of semi-supervised learning but also provide new insights into other areas, such as weakly supervised and unsupervised learning.

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